Tahlia Y. Wu

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EDUCATION

University of Pennsylvania, Philadelphia, PA - GPA: 4.0/4.0 (BSE), 4.0/4.0 (MSE) May 2028

Candidate for BSE in Mechanical Engineering, MSE in Robotics

Relevant Courses: Precision Machine Design and Manufacturing, Design of Mechatronic Systems, Thermodynamics, Statics, Dynamics, Solid Mechanics, Fluid Dynamics, Linear Algebra

EXPERIENCE

Mechatronics Engineer
Jan 2025-Present

Aware Lab, University of Pennsylvania, Philadelphia, PA

- Spearheaded **end-to-end mechatronics development** of the "FlowBot", a robotic traverse that automates data collection of anemometry data at 650+ 3D points across 20+ hours of fluid systems and wind tunnel experiments
- Achieved $\approx 6mm$ spatial positioning accuracy by programming real-time **PID closed-loop feedback control** integrated with OptiTrack motion capture telemetry
- Executed **mechanical assembly and electrical integration** for a custom 4-foot vertical linear actuator on a <$200 budget from off the shelf parts (e.g. 80/20 aluminum extrusions, stepper motors)
- Performed **structural and modal FEA** on vertical linear actuator to verify rigidity performance in surging unsteady aerodynamic conditions downstream of a wind turbine
- Engineered **multi-sensor data acquisition (DAQ) architectures**, speccing and integrating specialized hot-wire and sonic anemometers to increase temporal resolution to 20kHz
- Conducted **thermal analysis** on hotwire anemometer to determine overheat ratio for accurate sensor performance
- **Designed and manufactured 4 CAD-modeled components** for systems analysis And design of the motors, chassis, and vertical linear actuator
- Conducted rigorous torque calculations and technical **datasheet evaluations** to spec out components for integrating embedded electronics, motor drivers, and power distribution onto a mecanum chassis

Computational Air Quality Modeling Engineer
May 2025-Jul 2025

Department of Civil, Environmental and Geodetic Engineering, Ohio State University, Columbus, OH

- Engineered a scalable dispersion model in EPA MOVES/R-LINE, cutting simulation runtime to map nearside (<500m) roadside air pollution
- Parsed 2,000+ real-world road links via k-means into 5 speed bins and time periods, optimizing dataset performance
- Analyzed 12 vehicular pollutants, finding trucks emit up to 60.3% more NOx and 11.2x more PM2.5 than cars
- Formulated a superposition algorithm on a 50m unit line source, scaling model capability to 2,500m roads without reruns
- Mapped 36 complex roadways across 72 wind vectors to evaluate multi-intersection exposure in Whitehall, OH
- Presented air quality research findings at the Ohio State University Summer Symposium in front of 100+ attendees

Computer Vision R&D Engineer
Jun 2023-Oct 2023

Davis Lab of Computer Vision, UC Santa Cruz, Santa Cruz, CA

- Engineered a **3D perception and spatial AI pipeline** mapping point cloud topologies via 2D depth projection to enable 3D pose estimation and feature tracking for 468 dense landmarks.
- Optimized sensor processing and point cloud spatial alignment using kd-tree nearest-neighbor interpolation, filtering out visual noise to boost MediaPipe inference **reliability to 97.7%**.
- Developed **automated data preprocessing** pipelines integrating visibility algorithms and region of interest (ROI) cropping to resolve occlusions, back-projection errors, and out-of-bounds spatial artifacts.
- Achieved sub-centimeter-scale and **1.03±0.08% 3D coordinate error** in evaluation of pipeline's spatial localization precision across 129 3D scan topologies in the LYHM dataset.
- **Conducted model ablation and edge-case benchmarking** to prove spatial-only geometry processing eliminates dependency on visual texture data for real-time 3D perception systems.

PUBLICATIONS & PRESENTATIONS

Speaker *"Robotic Flow Measurements of Unsteady Wind-Turbine Wakes"* (APS Division of Fluid Dynamics, 2026)

2nd-Author *"A scalable framework for spatiotemporal modeling of traffic-related emissions"* (Submitted to ACS ES&T Air, 2026)

SKILLS

Software: Python, C, C++, Java, R, HTML/CSS, LabVIEW, Github, PyTorch, Computer Vision, MATLAB, Microsoft Office Suite

Robotics & Controls: PID Control, Closed-Loop Feedback, Kinematics, OptiTrack Motion Capture, Raspberry Pi, Arduino, PWM

CAD & Manufacturing Processes: SolidWorks (CSWA Certified), Mastercam, CNC Machining, Manual Mills & Lathes, 3D printing, Laser cutting, Sheet metal cutting, Tolerance stack-up, Soldering, DFM, Engineering Drawings, Bills of Materials (BOMs), GD&T